Project Proposal – Visual Analytics

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Project Title

Visual Analytics of AI Capability, Compute Infrastructure, and Governance Across Countries (2010–2024)

1. Dataset (Characteristics and Context)

This project integrates real, publicly available **country-level** datasets at **country–year** granularity.

Dataset scale

- **Entities:** 30 countries
- **Time range: 2010–2024 (15 years)**
- **Tuples:** 450 country–year observations (30 × 15)
- **Dimensions:** 23 analytical variables
- **Angelini–Santucci Index:**

$$AS = 450 \times 23 = 10{,}350$$

Data sources

- **World Bank Open Data:** GDP per capita, R&D expenditure, energy and ICT indicators
- **TOP500 Supercomputer List:** national high-performance computing capacity (count and FLOPs)
- **Kaggle datasets:** AI publications/citations/patents, cloud regions, data centers
- **OECD AI Policy Observatory:** national AI strategies and governance milestones
- **OECD / GPAI membership lists:** international coordination indicators

Variable groups (summary)

- **AI capability & innovation:** AI publications, citations, patents, R&D % GDP
- **AI compute & infrastructure (proxy indicators):** HPC capacity, cloud regions, data centers, energy/ICT indicators
- **AI governance & policy:** national AI strategy (binary/ordinal), regulation strictness score

- **International coordination & context:** OECD/GPAI membership, GDP per capita

Data preparation (planned)

- Harmonize country names and year coverage across sources
- Handle missing values using a consistent strategy (e.g., interpolation by year where appropriate, or removal of incomplete records)
- Standardize numeric variables (z-score) to support dimensionality reduction and clustering

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2. Objectives (Tasks and Analytical Questions)

The system supports exploratory analysis to reveal structural patterns in how countries differ in:

1. **Comparison / Ranking**

   - Which countries lead in compute capacity, AI research output, or policy readiness?
   - How large are cross-country gaps in a given year?

2. **Relationships and trade-offs**

   - How do compute indicators relate to governance strictness and innovation capacity?
   - Are there "mismatches" (e.g., high compute but weak governance, or strong governance with limited compute)?

3. **Similarity and clustering**

   - What "AI development profiles" emerge across countries?
   - Which countries are structural outliers?

4. **Temporal evolution**

   - How do national profiles change from 2010 to 2024?
   - Do countries move across clusters over time?

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3. Intended Users

The intended users are **policy analysts, researchers, and students** interested in global AI governance, compute infrastructure disparities, and international coordination. The system is designed for **data-driven exploratory investigation**, not for a single fixed narrative.

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4. Used Analytics

A) Dimensionality Reduction (PCA and/or UMAP)

Applied to the 23-dimensional country–year data to generate a **2D embedding** where proximity reflects similarity of multivariate profiles.

B) Clustering (k-means or density-based)

Used to identify groups of country–year observations with similar capability–compute–governance structures.

C) Temporal analysis

Used to track how a country's position in the embedded space changes over time (trajectories).

D) Mandatory analytics (interaction-triggered computation)

To satisfy the assignment requirement, the system will include **at least one computation triggered by user interaction**. Two candidate implementations:

- **Option 1 (recommended, robust): Selection summary statistics**
  When the user brushes/selects points in the embedding, the system computes and displays for the selected subset:
  - count of selected observations
  - mean/median/min/max of key variables (e.g., compute index, policy score, publications)
  - difference from a baseline (e.g., global mean or OECD mean)
- **Option 2 (advanced): "Ideal profile" distance-to-target**
  Users define a target profile (via sliders/weights for compute/policy/research), and the system computes the distance of each country–year to that target and recommends Top-K closest profiles with explanations.

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5. Visual Analytics Cycle Characteristics

- **Data:** Integration of heterogeneous real-world datasets
- **Analysis:** dimensionality reduction + clustering + temporal comparison
- **Visualization:** a coordinated multi-view dashboard
- **Interaction:** brushing, filtering, and selection trigger linked updates and analytic computations
- **Insight:** identification of structural roles, disparities, mismatches, and temporal shifts

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6. User Interface Design (Coordinated Multi-View System)

The dashboard will include **four coordinated views**, with two of them explicitly fulfilling the mandatory coordinated-visualization requirement.

(1) Main View — AI Development Landscape (required)

- **PCA/UMAP scatterplot** where each point is a country–year observation
- Color encoding: cluster membership or governance category
- Interactions: brushing (rectangular selection), clicking to pin a country, year filtering

(2) Coordinated View — Ranking / Comparison Bar Chart (required, bidirectional linking)

- A sorted bar chart ranking countries by a selected metric (compute index / policy score / research output)
- Clicking bars selects countries; multiple selection is supported
- **Bidirectional coordination (both ways):**
    - selection in the scatterplot highlights/filters the bar chart
    - selection in the bar chart highlights/filters points in the scatterplot

(3) Multivariate Profile View (explanation view)

- Parallel coordinates (preferred for 23 variables) showing the multivariate profile of selected countries/clusters
- Purpose: explain **why** a country belongs to a cluster or why it is an outlier

(4) Temporal Trajectory View (time evolution)

- A trajectory/path view showing how a selected country moves through the 2D embedding across years

- Supports comparison of trajectories across multiple countries

Control & Filtering Panel

- Year range slider
- Country/region filter
- Metric selector for ranking chart
- Optional: weights/targets for the “ideal profile” computation (if Option 2 is used)

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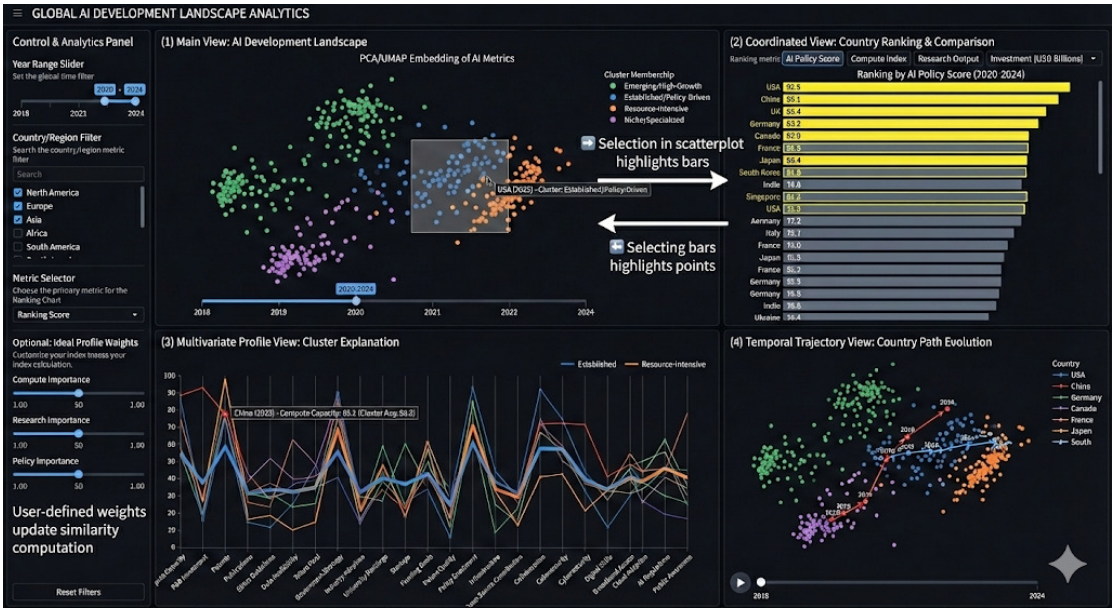
7. Visualization Design Rationale (Perception-Aware Choices)

- Primary quantitative comparisons rely on **position and length** (scatterplots and bar charts), avoiding misleading encodings such as 3D, angles, or area/volume for precise comparison.
- Color is used carefully:
  - distinct hues for **categorical** variables (region / governance category)
  - sequential (non-rainbow) scales for **quantitative** heatmaps (if used)
- Tooltips and clear labels provide context (units, year, sources) and reduce ambiguity.
- Below is the mockup of my user interface generated by Gemini



Conclusion

This project proposes an interactive visual analytics system that reveals structural patterns in global AI development by integrating measures of **capability**, **compute infrastructure**, and **governance**. The system explicitly satisfies the assignment requirements by providing:

1. **At least two interactive visualizations coordinated in both directions** (scatterplot ↔ ranking bar chart), and
2. **At least one analytic computation triggered by user interaction** (selection statistics and/or ideal-profile distance).

The outcome will support evidence-based exploratory analysis of disparities, clusters, governance–capacity mismatches, and temporal evolution across countries.